

```
[88]: import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
df=sns.load_dataset("penguins")
df.head()
```

```
[88]: species island bill_length_mm bill_depth_mm flipper_length_mm body_mass_g sex
0 Adeline Torgersen 39.1 18.7 181.0 3750.0 Male
1 Adeline Torgersen 39.5 17.4 186.0 3800.0 Female
2 Adeline Torgersen 40.3 18.0 195.0 3250.0 Female
3 Adeline Torgersen NaN NaN NaN NaN NaN
4 Adeline Torgersen 36.7 19.3 193.0 3450.0 Female
```

```
[89]: print("Descriptive statistics")
print(df.describe())

Descriptive statistics
bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
count 342.000000 342.000000 342.000000 342.000000
mean 43.921930 17.151170 200.915205 4201.754386
std 5.459584 1.974793 14.061714 801.954536
min 32.100000 13.100000 172.000000 2700.000000
25% 39.225000 15.600000 190.000000 3550.000000
50% 44.450000 17.300000 197.000000 4050.000000
75% 48.500000 18.700000 213.000000 4750.000000
max 59.600000 21.500000 231.000000 6300.000000
```

```
[90]: print("Skewness Values")
numeric_cols=df.select_dtypes(include=np.number)
print(numeric_cols.skew())

Skewness Values
bill_length_mm 0.053118
bill_depth_mm -0.143465
flipper_length_mm 0.345682
body_mass_g 0.470329
dtype: float64
```

```
[91]: #apply z-score
zscore_df=numeric_cols.copy()
for col in zscore_df.columns:
    zscore_df[col]=(
        zscore_df[col]-zscore_df[col].mean()
    )/zscore_df.std()
print(zscore_df.head())

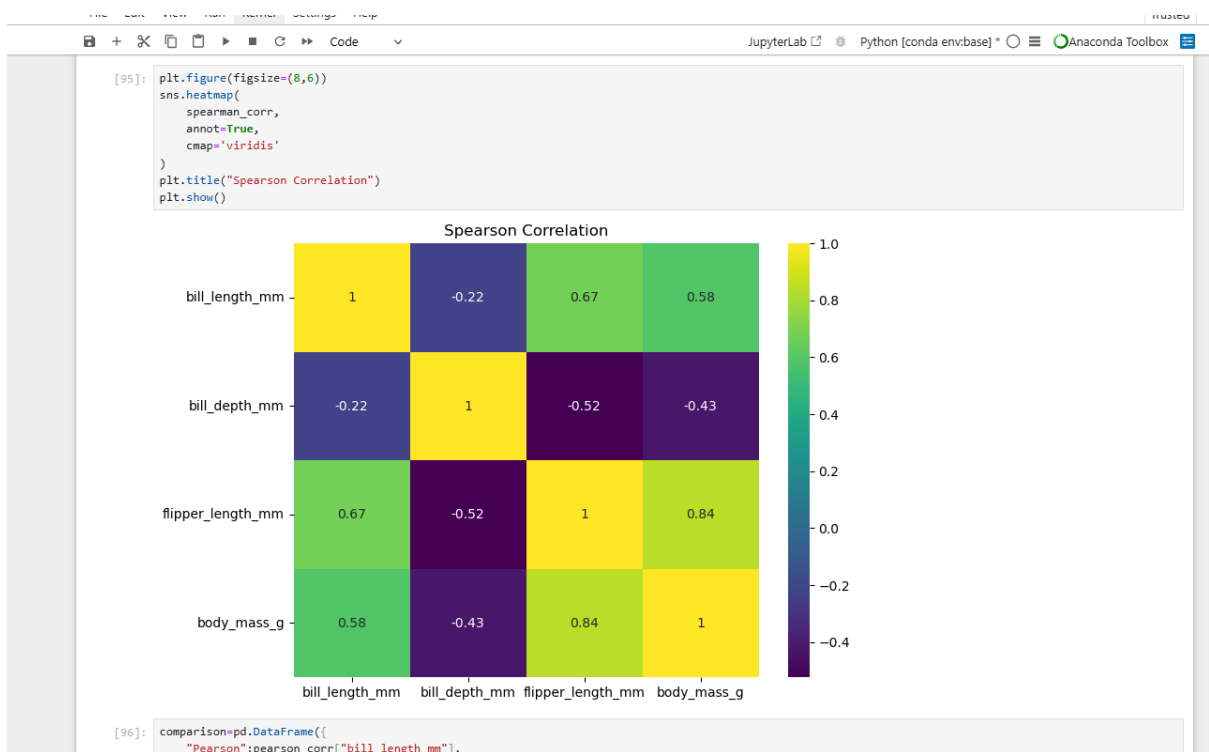
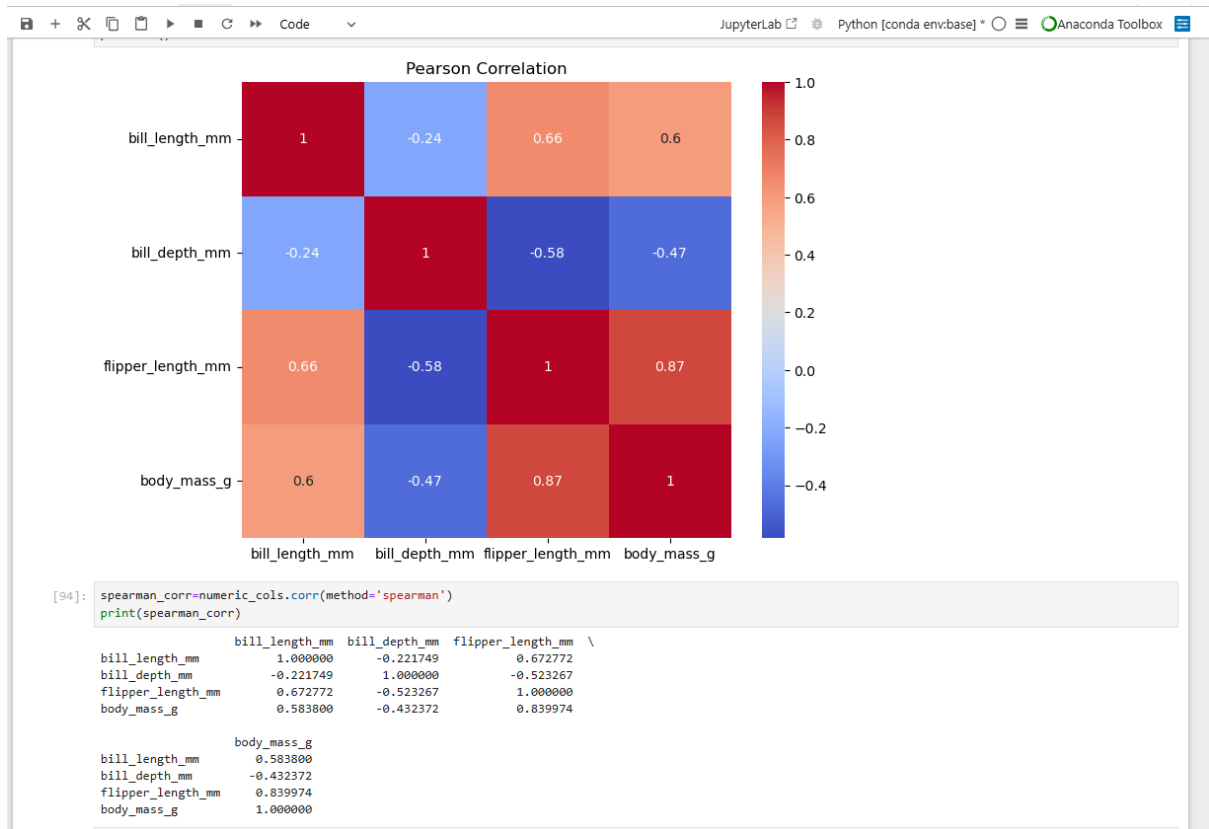
bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
0 NaN NaN NaN NaN
1 NaN NaN NaN NaN
2 NaN NaN NaN NaN
3 NaN NaN NaN NaN
4 NaN NaN NaN NaN
```

```
[92]: pearson_corr = numeric_cols.corr(method='pearson')
print(pearson_corr)

bill_length_mm bill_depth_mm flipper_length_mm body_mass_g \
bill_length_mm 1.000000 -0.235053 0.656181
bill_depth_mm -0.235053 1.000000 -0.583851
flipper_length_mm 0.656181 -0.583851 1.000000
body_mass_g 0.595110 -0.471916 0.871202

body_mass_g
bill_length_mm 0.595110
bill_depth_mm -0.471916
flipper_length_mm 0.871202
body_mass_g 1.000000
```

```
[93]: plt.figure(figsize=(8,6))
sns.heatmap(
    pearson_corr,
    annot=True,
    cmap='coolwarm'
)
plt.title("Pearson Correlation")
plt.show()
```



```
File Edit View Run Kernel Settings Help Trusted
JupyterLab Python [conda env:base] Anaconda Toolbox

[96]: comparison=pd.DataFrame({
      "Pearson":pearson_corr["bill_length_mm"],
      "Spearman":spearman_corr["bill_length_mm"]
    })
      print(comparison)

      Pearson Spearman
bill_length_mm 1.000000 1.000000
bill_depth_mm -0.235053 -0.221749
flipper_length_mm 0.656181 0.672772
body_mass_g 0.595110 0.583800

[97]: df=sns.load_dataset("titanic")
      print(df.head())
      print(df.shape)

      survived pclass sex age sibsp parch fare embarked class \
0 0 3 male 22.0 1 0 7.2500 S Third
1 1 1 female 38.0 1 0 71.2833 C First
2 1 3 female 26.0 0 0 7.9250 S Third
3 1 1 female 35.0 1 0 53.1000 S First
4 0 3 male 35.0 0 0 8.0500 S Third

      who adult_male deck embark_town alive alone
0 man True NaN Southampton no False
1 woman False C Cherbourg yes False
2 woman False NaN Southampton yes True
3 woman False C Southampton yes False
4 man True NaN Southampton no True
(891, 15)

[98]: df.loc[0,'embark_town']='southampton'
      df.loc[1,'embark_town']='Southampton'
      df.loc[2,'embark_town']='SOUTHAMPTON'
      print(df['embark_town'].unique())

      ['southampton' 'Southampton' 'SOUTHAMPTON' 'Queenstown' 'Cherbourg' nan]

[99]: df['embark_town']=df['embark_town'].str.title()

[100]: print(df['embark_town'].unique())

      ['Southampton' 'Queenstown' 'Cherbourg' nan]

[101]: print(df['embark_town'].isnull().sum())

      2
```

```
File Edit View Run Kernel Settings Help Trusted
JupyterLab Python [conda env:base] Anaconda Toolbox

[102]: df['embark_town']=df['embark_town'].fillna('Kingstown')
      print(df['embark_town'].unique())

      ['Southampton' 'Queenstown' 'Cherbourg' 'Kingstown']

[103]: duplicate_rows=df.iloc[:5]
      df_dup=pd.concat([df,duplicate_rows])
      print(df_dup.shape)

      (896, 15)

[104]: print(df_dup.duplicated().sum())

      112

[105]: df_dup[df_dup.duplicated()]

      survived pclass sex age sibsp parch fare embarked class who adult_male deck embark_town alive alone
47 1 3 female NaN 0 0 7.7500 Q Third woman False NaN Queenstown yes True
76 0 3 male NaN 0 0 7.8958 S Third man True NaN Southampton no True
77 0 3 male NaN 0 0 8.0500 S Third man True NaN Southampton no True
87 0 3 male NaN 0 0 8.0500 S Third man True NaN Southampton no True
95 0 3 male NaN 0 0 8.0500 S Third man True NaN Southampton no True
... ... ... ... ... ... ... ... ... ...
0 0 3 male 22.0 1 0 7.2500 S Third man True NaN Southampton no False
1 1 1 female 38.0 1 0 71.2833 C First woman False C Southampton yes False
2 1 3 female 26.0 0 0 7.9250 S Third woman False NaN Southampton yes True
3 1 1 female 35.0 1 0 53.1000 S First woman False C Southampton yes False
4 0 3 male 35.0 0 0 8.0500 S Third man True NaN Southampton no True

      112 rows x 15 columns

[106]: df_dup=df_dup.drop_duplicates()
      print(df_dup.duplicated().sum())

      0
```

```
[107]: print(df['embark_town'].value_counts())
```

```
embark_town
Southampton    645
Cherbourg      167
Queenstown     77
Kingstown       2
Name: count, dtype: int64
```

```
[108]: df.loc[0,'embark_town']='Southamton'
df.loc[1,'embark_town']='Southhampton'
```

```
[109]: df.loc[0,'embark_town']='Southamton'
df.loc[1,'embark_town']='Southhamton'
print(df['embark_town'].value_counts())
```

```
embark_town
Southampton    643
Cherbourg      167
Queenstown     77
Kingstown       2
Southamton      1
Southhamton     1
Name: count, dtype: int64
```

```
[110]: #Fix Typos
df['embark_town']=df['embark_town'].replace({
    'Southamton':'Southampton',
    'Southhampton':'Sothhampton'
})
print(df['embark_town'].value_counts())
```

```
embark_town
Southampton    644
Cherbourg      167
Queenstown     77
Kingstown       2
Southhamton     1
Name: count, dtype: int64
```

```
[111]: #detect possible typos
print(sorted(df['embark_town'].dropna().unique()))

['Cherbourg', 'Kingstown', 'Queenstown', 'Southampton', 'Southhamton']
```

```
[112]: df.loc[0,'age']=240
```

```
[113]: #Detect Unusual values
df['age'].describe()
```

```
[113]: count    714.000000
      mean     30.371387
      std     19.362012
      min      0.420000
      25%     20.125000
      50%     28.000000
      75%     38.750000
      max     300.000000
      Name: age, dtype: float64
```

```
[114]: #Fix unit mismatch
#convert months to years
df.loc[df['age']>120,'age']=(
    df.loc[df['age']>120,'age']/12
)
df['age'].describe()
```

```
[114]: count    714.000000
      mean     29.678109
      std     14.525902
      min      0.420000
      25%     20.000000
      50%     28.000000
      75%     38.000000
      max      80.000000
      Name: age, dtype: float64
```

```
[115]: df['survived_label']=df['survived'].map({
      0:'No',
      1:'Yes'
})
print(df.head())
```

```
   survived  pclass    sex  age  sibsp  parch   fare embarked  class \
0         0      3  male  20.0    1     0   7.2500      S  Third
1         1      1 female  25.0    1     0  71.2833      C  First
2         1      3 female  26.0    0     0   7.9250      S  Third
3         1      1 female  35.0    1     0  53.1000      S  First
4         0      3  male  35.0    0     0   8.0500      S  Third
```

```
   who  adult_male  deck  embark_town  alive  alone  survived_label
0  man           True   NaN  Southampton    no  False             No
```

```
[116]: df.drop(columns=['deck'],inplace=True)
df.head()
```

```
[116]:   survived  pclass    sex  age  sibsp  parch   fare  embarked  class  who  adult_male  embark_town  alive  alone  survived_label
0         0      3  male  20.0    1     0   7.2500      S  Third  man           True  Southampton    no  False             No
1         1      1 female  25.0    1     0  71.2833      C  First woman          False  Southampton    yes  False             Yes
2         1      3 female  26.0    0     0   7.9250      S  Third woman          False  Southampton    yes  True             Yes
3         1      1 female  35.0    1     0  53.1000      S  First woman          False  Southampton    yes  False             Yes
4         0      3  male  35.0    0     0   8.0500      S  Third  man           True  Southampton    no  True             No
```

```
[117]: #filling missing values
```

```
#1.Mean imputation
#replace missing values with avg value
df['age']=df['age'].fillna(df['age'].mean())
df.head(20)
```

```
[117]:   survived  pclass    sex  age  sibsp  parch   fare  embarked  class  who  adult_male  embark_town  alive  alone  survived_label
0         0      3  male  20.000000    1     0   7.2500      S  Third  man           True  Southampton    no  False             No
1         1      1 female  25.000000    1     0  71.2833      C  First woman          False  Southampton    yes  False             Yes
2         1      3 female  26.000000    0     0   7.9250      S  Third woman          False  Southampton    yes  True             Yes
3         1      1 female  35.000000    1     0  53.1000      S  First woman          False  Southampton    yes  False             Yes
4         0      3  male  35.000000    0     0   8.0500      S  Third  man           True  Southampton    no  True             No
5         0      3  male  29.678109    0     0   8.4583      Q  Third  man           True  Queenstown    no  True             No
6         0      1  male  54.000000    0     0  51.8625      S  First  man           True  Southampton    no  True             No
7         0      3  male   2.000000    3     1  21.0750      S  Third  child          False  Southampton    no  False             No
8         1      3 female  27.000000    0     2  11.1333      S  Third woman          False  Southampton    yes  False             Yes
9         1      2 female  14.000000    1     0  30.0708      C  Second  child          False  Cherbourg    yes  False             Yes
10        1      3 female   4.000000    1     1  16.7000      S  Third  child          False  Southampton    yes  False             Yes
11        1      1 female  58.000000    0     0  26.5500      S  First woman          False  Southampton    yes  True             Yes
```

```
[118]: #2.Median Imputation
```

```
df['age']=df['age'].fillna(df['age'].median())
df.head(20)
```

```
[118]:
```

|    | survived | pclass | sex    | age       | sibsp | parch | fare    | embarked | class  | who   | adult_male | embark_town | alive | alone | survived_label |
|----|----------|--------|--------|-----------|-------|-------|---------|----------|--------|-------|------------|-------------|-------|-------|----------------|
| 0  | 0        | 3      | male   | 20.000000 | 1     | 0     | 7.2500  | S        | Third  | man   | True       | Southampton | no    | False | No             |
| 1  | 1        | 1      | female | 25.000000 | 1     | 0     | 71.2833 | C        | First  | woman | False      | Southampton | yes   | False | Yes            |
| 2  | 1        | 3      | female | 26.000000 | 0     | 0     | 7.9250  | S        | Third  | woman | False      | Southampton | yes   | True  | Yes            |
| 3  | 1        | 1      | female | 35.000000 | 1     | 0     | 53.1000 | S        | First  | woman | False      | Southampton | yes   | False | Yes            |
| 4  | 0        | 3      | male   | 35.000000 | 0     | 0     | 8.0500  | S        | Third  | man   | True       | Southampton | no    | True  | No             |
| 5  | 0        | 3      | male   | 29.678109 | 0     | 0     | 8.4583  | Q        | Third  | man   | True       | Queenstown  | no    | True  | No             |
| 6  | 0        | 1      | male   | 54.000000 | 0     | 0     | 51.8625 | S        | First  | man   | True       | Southampton | no    | True  | No             |
| 7  | 0        | 3      | male   | 2.000000  | 3     | 1     | 21.0750 | S        | Third  | child | False      | Southampton | no    | False | No             |
| 8  | 1        | 3      | female | 27.000000 | 0     | 2     | 11.1333 | S        | Third  | woman | False      | Southampton | yes   | False | Yes            |
| 9  | 1        | 2      | female | 14.000000 | 1     | 0     | 30.0708 | C        | Second | child | False      | Cherbourg   | yes   | False | Yes            |
| 10 | 1        | 3      | female | 4.000000  | 1     | 1     | 16.7000 | S        | Third  | child | False      | Southampton | yes   | False | Yes            |
| 11 | 1        | 1      | female | 58.000000 | 0     | 0     | 26.5500 | S        | First  | woman | False      | Southampton | yes   | True  | Yes            |
| 12 | 0        | 3      | male   | 20.000000 | 0     | 0     | 8.0500  | S        | Third  | man   | True       | Southampton | no    | True  | No             |
| 13 | 0        | 3      | male   | 39.000000 | 1     | 5     | 31.2750 | S        | Third  | man   | True       | Southampton | no    | False | No             |
| 14 | 0        | 3      | female | 14.000000 | 0     | 0     | 7.8542  | S        | Third  | child | False      | Southampton | no    | True  | No             |
| 15 | 1        | 2      | female | 55.000000 | 0     | 0     | 16.0000 | S        | Second | woman | False      | Southampton | yes   | True  | Yes            |
| 16 | 0        | 3      | male   | 2.000000  | 4     | 1     | 29.1250 | Q        | Third  | child | False      | Queenstown  | no    | False | No             |
| 17 | 1        | 2      | male   | 29.678109 | 0     | 0     | 13.0000 | S        | Second | man   | True       | Southampton | yes   | True  | Yes            |
| 18 | 0        | 3      | female | 31.000000 | 1     | 0     | 18.0000 | S        | Third  | woman | False      | Southampton | no    | False | No             |
| 19 | 1        | 3      | female | 29.678109 | 0     | 0     | 7.2250  | C        | Third  | woman | False      | Cherbourg   | yes   | True  | Yes            |

```
[119]: #Mode Imputation
```

```
df['age']=df['age'].fillna(df['age'].mode())
```

```
[119]: #Mode Imputation
```

```
df['age']=df['age'].fillna(df['age'].mode())
df.head(20)
```

```
[119]:
```

|    | survived | pclass | sex    | age       | sibsp | parch | fare    | embarked | class  | who   | adult_male | embark_town | alive | alone | survived_label |
|----|----------|--------|--------|-----------|-------|-------|---------|----------|--------|-------|------------|-------------|-------|-------|----------------|
| 0  | 0        | 3      | male   | 20.000000 | 1     | 0     | 7.2500  | S        | Third  | man   | True       | Southampton | no    | False | No             |
| 1  | 1        | 1      | female | 25.000000 | 1     | 0     | 71.2833 | C        | First  | woman | False      | Southampton | yes   | False | Yes            |
| 2  | 1        | 3      | female | 26.000000 | 0     | 0     | 7.9250  | S        | Third  | woman | False      | Southampton | yes   | True  | Yes            |
| 3  | 1        | 1      | female | 35.000000 | 1     | 0     | 53.1000 | S        | First  | woman | False      | Southampton | yes   | False | Yes            |
| 4  | 0        | 3      | male   | 35.000000 | 0     | 0     | 8.0500  | S        | Third  | man   | True       | Southampton | no    | True  | No             |
| 5  | 0        | 3      | male   | 29.678109 | 0     | 0     | 8.4583  | Q        | Third  | man   | True       | Queenstown  | no    | True  | No             |
| 6  | 0        | 1      | male   | 54.000000 | 0     | 0     | 51.8625 | S        | First  | man   | True       | Southampton | no    | True  | No             |
| 7  | 0        | 3      | male   | 2.000000  | 3     | 1     | 21.0750 | S        | Third  | child | False      | Southampton | no    | False | No             |
| 8  | 1        | 3      | female | 27.000000 | 0     | 2     | 11.1333 | S        | Third  | woman | False      | Southampton | yes   | False | Yes            |
| 9  | 1        | 2      | female | 14.000000 | 1     | 0     | 30.0708 | C        | Second | child | False      | Cherbourg   | yes   | False | Yes            |
| 10 | 1        | 3      | female | 4.000000  | 1     | 1     | 16.7000 | S        | Third  | child | False      | Southampton | yes   | False | Yes            |
| 11 | 1        | 1      | female | 58.000000 | 0     | 0     | 26.5500 | S        | First  | woman | False      | Southampton | yes   | True  | Yes            |
| 12 | 0        | 3      | male   | 20.000000 | 0     | 0     | 8.0500  | S        | Third  | man   | True       | Southampton | no    | True  | No             |
| 13 | 0        | 3      | male   | 39.000000 | 1     | 5     | 31.2750 | S        | Third  | man   | True       | Southampton | no    | False | No             |
| 14 | 0        | 3      | female | 14.000000 | 0     | 0     | 7.8542  | S        | Third  | child | False      | Southampton | no    | True  | No             |
| 15 | 1        | 2      | female | 55.000000 | 0     | 0     | 16.0000 | S        | Second | woman | False      | Southampton | yes   | True  | Yes            |
| 16 | 0        | 3      | male   | 2.000000  | 4     | 1     | 29.1250 | Q        | Third  | child | False      | Queenstown  | no    | False | No             |
| 17 | 1        | 2      | male   | 29.678109 | 0     | 0     | 13.0000 | S        | Second | man   | True       | Southampton | yes   | True  | Yes            |
| 18 | 0        | 3      | female | 31.000000 | 1     | 0     | 18.0000 | S        | Third  | woman | False      | Southampton | no    | False | No             |
| 19 | 1        | 3      | female | 29.678109 | 0     | 0     | 7.2250  | C        | Third  | woman | False      | Cherbourg   | yes   | True  | Yes            |

## KNN Imputation

```
[124]: import pandas as pd
import numpy as np
data={
    'age':[22,25,np.nan,23,24],
    'fare':[7,28,8.09,8.03,7.9]
}
df=pd.DataFrame(data)
print(df)
```

```
   age  fare
0  22.0   7.00
1  25.0  28.00
2   NaN   8.09
3  23.0   8.03
4  24.0   7.90
```

```
[127]: from sklearn.impute import KNNImputer
KNN=KNNImputer(n_neighbors=3)
df_KNN=pd.DataFrame(
    KNN.fit_transform(df),
    columns=df.columns)
print(df_KNN)
```

```
   age  fare
0  22.0   7.00
1  25.0  28.00
2  23.0   8.09
3  23.0   8.03
4  24.0   7.90
```

```
[129]: df=sns.load_dataset('titanic')
df['embarked_ffill']=df['embarked'].ffill()
df.head()
```

```
[129]:
```

|   | survived | pclass | sex    | age  | sibsp | parch | fare    | embarked | class | who   | adult_male | deck | embark_town | alive | alone | embarked_ffill |
|---|----------|--------|--------|------|-------|-------|---------|----------|-------|-------|------------|------|-------------|-------|-------|----------------|
| 0 | 0        | 3      | male   | 22.0 | 1     | 0     | 7.2500  | S        | Third | man   | True       | NaN  | Southampton | no    | False | S              |
| 1 | 1        | 1      | female | 38.0 | 1     | 0     | 71.2833 | C        | First | woman | False      | C    | Cherbourg   | yes   | False | C              |
| 2 | 1        | 3      | female | 26.0 | 0     | 0     | 7.9250  | S        | Third | woman | False      | NaN  | Southampton | yes   | True  | S              |
| 3 | 1        | 1      | female | 35.0 | 1     | 0     | 53.1000 | S        | First | woman | False      | C    | Southampton | yes   | False | S              |

```
[130]: cols=[
    'age',
    'fare',
    'sibsp',
    'parch',
    'pclass'
]
imputer=KNNImputer(n_neighbors=5)
df[cols]=imputer.fit_transform(df[cols])
print(df.isnull().sum())
```

```
survived      0
pclass        0
sex           0
age           0
sibsp         0
parch         0
fare          0
embarked      2
class         0
who           0
adult_male    0
deck         688
embark_town   2
alive         0
alone         0
embarked_ffill 0
dtype: int64
```

```
[131]: import pandas as pd
from sklearn.preprocessing import LabelEncoder
import seaborn as sns
df=sns.load_dataset("titanic")
le=LabelEncoder()
df['sex_encoded']=le.fit_transform(
    df['sex']
)

# one hot encoding
df=pd.get_dummies(
    df,
    columns=['embarked']
)
```

```
# one hot encoding
df=pd.get_dummies(
    df,
    columns=['embarked']
)

#target encoding
target_map=df.groupby(
    'pclass'
)['survived'].mean()

df['class_target']=(
    df['pclass']
    .map(target_map)
)
print(df.head())
```

|   | survived | pclass | sex    | age  | sibsp | parch | fare    | class | who   | \ |
|---|----------|--------|--------|------|-------|-------|---------|-------|-------|---|
| 0 | 0        | 3      | male   | 22.0 | 1     | 0     | 7.2500  | Third | man   |   |
| 1 | 1        | 1      | female | 38.0 | 1     | 0     | 71.2833 | First | woman |   |
| 2 | 1        | 3      | female | 26.0 | 0     | 0     | 7.9250  | Third | woman |   |
| 3 | 1        | 1      | female | 35.0 | 1     | 0     | 53.1000 | First | woman |   |
| 4 | 0        | 3      | male   | 35.0 | 0     | 0     | 8.0500  | Third | man   |   |

|   | adult_male | deck | embark_town | alive | alone | sex_encoded | embarked_C | \ |
|---|------------|------|-------------|-------|-------|-------------|------------|---|
| 0 | True       | NaN  | Southampton | no    | False | 1           | False      |   |
| 1 | False      | C    | Cherbourg   | yes   | False | 0           | True       |   |
| 2 | False      | NaN  | Southampton | yes   | True  | 0           | False      |   |
| 3 | False      | C    | Southampton | yes   | False | 0           | False      |   |
| 4 | True       | NaN  | Southampton | no    | True  | 1           | False      |   |

|   | embarked_Q | embarked_S | class_target |
|---|------------|------------|--------------|
| 0 | False      | True       | 0.242363     |
| 1 | False      | False      | 0.629630     |
| 2 | False      | True       | 0.242363     |
| 3 | False      | True       | 0.629630     |
| 4 | False      | True       | 0.242363     |

[ ]: imp